Multi-Label Lung Disease Classification Using EfficientNet-B1

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Abstract. Deep learning approaches have demonstrated significant potential in automated chest X-ray analysis, and this study presents a comprehensive multi-label classification framework for detecting 12 lungs diseases using the NIH ChestX-ray14 dataset, addressing critical class imbalance through strategic dataset balancing and oversampling techniques while employing advanced data augmentation methods including random horizontal flipping, rotational transformations, and color jittering to enhance model generalization. Our implementation incorporates multiple regularization strategies such as dropout layers and L2 weight decay to prevent overfitting, utilizes Binary Cross-Entropy loss with Adam optimization for effective multi-label learning, and employs a comprehensive evaluation methodology encompassing AUC-ROC curves, F1-scores, precision, and recall with class-specific threshold optimization to maximize diagnostic performance. The proposed solution demonstrates that carefully engineered deep learning pipelines with appropriate preprocessing, augmentation, and regularization techniques can achieve strong performance in medical image classification, highlighting the importance of addressing dataset imbalances and employing robust training methodologies for effective clinical decision support systems in radiology.

Keywords: deep learning, chest X-ray classification, multi-label classification, medical image analysis.

1. Introduction

Chest X-rays remain one of the most widely used and cost-effective medical imaging modalities for diagnosing thoracic diseases, serving as a primary tool for detecting conditions ranging from pneumonia and lung nodules to pleural effusions and pneumothorax. However, the increasing volume of radiographic examinations, coupled with the shortage of specialized radiologists in many regions, has created significant challenges in healthcare systems worldwide. This diagnostic bottleneck has accelerated the need for automated computer-aided diagnosis systems that can assist medical professionals in interpreting chest radiographs accurately and efficiently.

The emergence of deep learning technologies has revolutionized medical image analysis, offering unprecedented opportunities for automated disease detection and classification. Particularly in chest X-ray interpretation, convolutional neural networks have demonstrated remarkable capabilities in identifying multiple pathologies simultaneously from single images. However, several significant challenges persist in developing robust classification systems, including the inherent class imbalance in medical datasets where certain diseases appear far less frequently than others, the complexity of multi-label classification where multiple conditions can coexist in a single patient, and the need for models that generalize well across diverse patient populations and imaging conditions.

This research addresses these challenges through a comprehensive framework for multi-label thoracic disease classification, leveraging the NIH ChestX-ray14 dataset to develop and validate an automated diagnosis system. Our work focuses on implementing sophisticated data balancing techniques to mitigate class imbalance issues, employing advanced augmentation strategies to enhance model generalization, and incorporating multiple regularization methods to prevent overfitting while maintaining high diagnostic accuracy. By optimizing the entire pipeline from data preprocessing to model evaluation, we aim to contribute to the development of reliable decision support tools that can assist radiologists in clinical practice, potentially improving diagnostic accuracy and reducing interpretation times for better patient outcomes.

2. Related work

The application of deep learning to chest X-ray analysis has advanced substantially since the creation of the NIH ChestX-ray14 dataset [1], which provided a large-scale benchmark for thoracic disease classification. Early work by Yao et al. [2] demonstrated the value of recurrent neural networks for capturing dependencies between multiple diagnostic labels. Rajpurkar et al. [3] further advanced the field with CheXNet, a deep convolutional network that achieved expert-level performance in detecting pneumonia. A persistent challenge in this domain has been class imbalance, which Johnson and Khoshgoftaar [4] systematically addressed through comparative analysis of sampling techniques. Buda et al. [5] provided additional evidence that rebalancing strategies significantly improve recognition of rare conditions, while Wang et al. [6] showed similar benefits from loss re-weighting approaches. Transfer learning has emerged as another critical component, with Tajbakhsh et al. [7] establishing that pretrained models consistently outperform training from scratch. Raghu et al. [8] validated these findings across diverse network architectures, and Tan and Le [9] introduced EfficientNet, which later demonstrated strong performance on medical imaging tasks as shown by Pham et al. [10]. Data augmentation has proven equally important, with Shorten and Khoshgoftaar [11] documenting the benefits of geometric and photometric transformations. Chlap et al. [12] specialized these techniques for medical imaging contexts, and Chen et al. [13] developed chest-specific augmentation methods. Multi-label classification presents unique challenges that Li et al. [14] addressed through label correlation models, while Liu et al. [15] employed graph networks to explicitly

represent disease co-occurrence patterns. Wang et al. [16] incorporated attention mechanisms to improve both localization and classification. Regularization techniques have been crucial for generalization, beginning with dropout introduced by Srivastava et al. [17] and extended through weight decay optimization by Loshchilov and Hutter [18]. Touvron et al. [19] demonstrated these methods' importance for medical data specifically. Evaluation methodologies have likewise evolved, with DeLong et al. [20] establishing robust ROC comparison methods and Saito and Rehmsmeier [21] highlighting precision-recall analysis for imbalanced data. Recent guidelines from Mongan et al. [22] have standardized evaluation protocols across the field. Our work integrates these diverse advances into a unified framework for multi-label thoracic disease classification.

3. Methodology

3.1. Dataset

The study utilizes the NIH ChestX-ray14 dataset, one of the largest publicly available collections of chest radiographs containing frontal-view X-ray images with 14 disease labels. For this research, we work with a balanced subset comprising 13 thoracic conditions: Atelectasis, Effusion, Infiltration, Mass, Nodule, Pneumonia, Pneumothorax, Consolidation, Edema, Emphysema, Fibrosis, Pleural_Thickening, and No Finding. The dataset is strategically partitioned into training, validation, and test sets with 29,382 images in total, ensuring representative distribution across all pathological conditions.

To address the significant class imbalance inherent in medical imaging datasets, we implement an intelligent oversampling strategy during training. Each disease class is balanced to contain exactly 2000 samples through augmentation of minority classes, while majority classes are randomly downsampled to prevent model bias toward frequently occurring conditions. This balanced approach ensures equal learning opportunity for all pathologies regardless of their natural prevalence in the dataset.

The dataset preprocessing pipeline incorporates comprehensive quality checks and file validation to handle missing or corrupted images. Each image is standardized to 224×224 pixel resolution and normalized using ImageNet statistics (mean: [0.485, 0.456, 0.406], std: [0.229, 0.224, 0.225]) to maintain consistency with pretrained model expectations. The multi-label annotation format preserves the clinical reality that multiple thoracic conditions can coexist in a single patient, requiring specialized learning approaches beyond conventional single-label classification.

Pathology Label	Count	After Augmentation
Atelectasis	3235	2000
Effusion	3160	2000
Infiltration	3327	2000
Mass	1943	2000
Nodule	2163	2000
Pneumonia	1400	2000
Pneumothorax	2290	2000
Consolidation	2067	2000
Edema	1738	2000
Emphysema	1463	2000
Fibrosis	1411	2000
Pleural Thickening	1607	2000
No findings	3578	2000

Table1: Dataset statistics

3.2. Regularisation

To mitigate overfitting and enhance generalization on the medical imaging task, our approach incorporates a comprehensive regularization strategy combining multiple techniques. We employ architectural regularization through dropout layers ($p=0.2$) in the classifier head to prevent neuron co-adaptation, complemented by L2 weight decay ($\lambda=1e-4$) in the Adam optimizer to constrain model complexity and prevent parameter explosion. Data-level regularization is achieved through extensive augmentation including random horizontal flipping, rotational transformations, and color jittering, alongside strategic class balancing that oversamples minority classes to ensure equal learning opportunity. These methods are further reinforced by early stopping based on validation AUC monitoring and the inherent batch normalization from the pre-trained EfficientNet backbone, collectively creating a robust framework that maintains performance on unseen medical data while preventing overfitting to the training distribution.

3.3. Loss function

Our multi-label chest X-ray classification framework employs Binary Cross-Entropy (BCE) Loss to effectively handle the complex medical scenario where patients can present with multiple coexisting thoracic conditions simultaneously. The BCE loss

computes independent binary classification losses for each of the 13 thoracic diseases and averages them, working in conjunction with sigmoid activation in the final layer to produce probability estimates between 0-1 for each pathology. This formulation maintains stable gradient propagation during training while naturally accommodating the multi-label nature of chest X-ray diagnosis, where a single image may contain evidence of several conditions that require independent detection and classification without mutual exclusion constraints.

3.4. Architecture

Our model architecture builds upon the EfficientNet-B1 backbone pre-trained on ImageNet, leveraging its proven capability in feature extraction while adapting it for multi-label medical image classification. The base convolutional neural network serves as a feature extractor, followed by a custom classifier head comprising a dropout layer (p=0.2) for regularization, a linear layer that maps 1280 features to 13 output nodes corresponding to our thoracic disease classes, and a sigmoid activation function that enables independent multi-label probability estimation. This architectural design harnesses the power of transfer learning while specifically optimizing for the complex task of detecting multiple coexisting pathologies in chest X-rays through parallel binary classification pathways.

3.5. Training Procedure

The model was trained using a comprehensive pipeline designed for multi-label medical image classification. The training employed the Adam optimizer with a learning rate of 0.001 and weight decay of 1e-4, using Binary Cross-Entropy loss to handle multiple simultaneous pathologies. The dataset was divided into training, validation, and test splits, with the training set balanced through oversampling to achieve 1000 samples per class. During each of the 20 training epochs, the model processed data in batches of 32 images, with extensive data augmentation including random horizontal flipping, rotation, and color jittering applied exclusively to training samples. Validation was performed after each epoch to monitor performance, with model checkpointing based on validation AUC to implement early stopping. The training incorporated multiple regularization techniques including dropout (p=0.2) and L2 weight decay to prevent overfitting, while maintaining the original, unaugmented validation and test sets for reliable performance evaluation.

4. Results

4.1. Evaluation Metrics

The model performance was comprehensively evaluated using multiple metrics tailored for multi-label medical image classification. The primary evaluation metric was the Area Under the Receiver Operating Characteristic Curve (AUC-ROC), calculated

independently for each of the 13 thoracic disease classes to assess the model's discrimination ability across all pathologies. Additionally, precision, recall, and F1-score were computed for each class using optimal thresholding, where thresholds between 0.1 and 0.5 were systematically evaluated to maximize F1-scores per disease. The mean values across all classes provided aggregate performance measures, while per-class metrics ensured detailed assessment of model performance on both common and rare conditions. This multi-faceted evaluation approach enabled thorough analysis of the model's diagnostic capabilities while accounting for the class imbalance inherent in medical datasets.

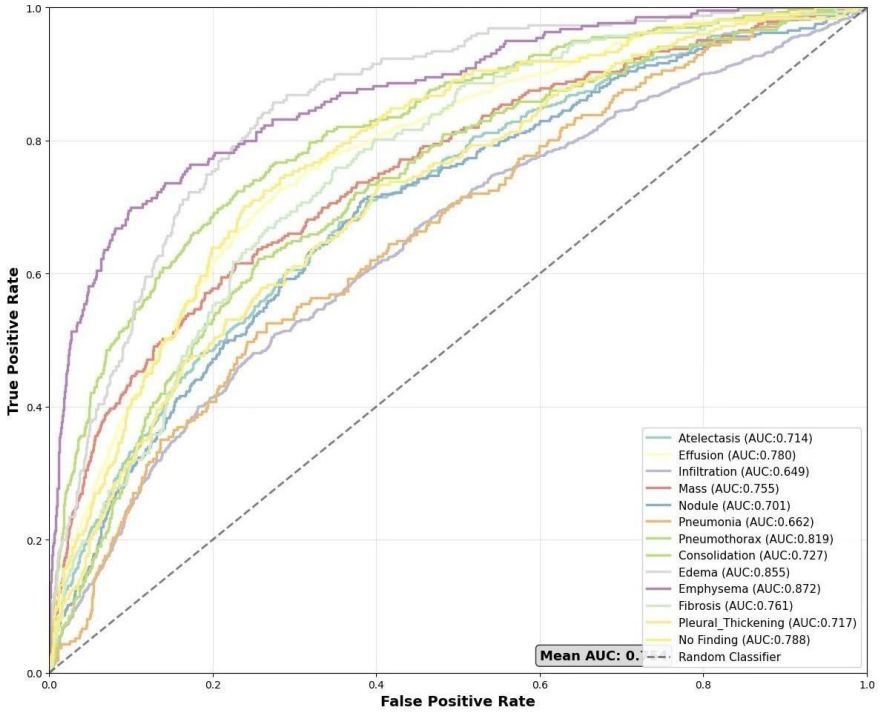


Figure 1: AUC-ROC Curve

Each curve represents the trade-off between true positive rate and false positive rate for a specific disease, along with its corresponding AUC value. The model demonstrates moderate to strong discriminative performance across most classes, with an overall mean AUC of approximately 0.75.

Pathology Label	AUC	Precision	Recall	F1-Score
Atelectasis	0.714	0.384	0.424	0.403
Effusion	0.78	0.465	0.667	0.548
Infiltration	0.649	0.368	0.762	0.496
Mass	0.755	0.334	0.45	0.383
Nodule	0.701	0.26	0.414	0.319
Pneumonia	0.662	0.113	0.645	0.192
Pneumothorax	0.819	0.449	0.515	0.479
Consolidation	0.727	0.281	0.275	0.278
Edema	0.855	0.334	0.619	0.434
Emphysema	0.872	0.625	0.5	0.556
Fibrosis	0.761	0.17	0.67	0.271
Pleural Thickening	0.717	0.183	0.579	0.279
Average	0.75	0.321	0.548	0.381

Table 2: Evaluation metrices

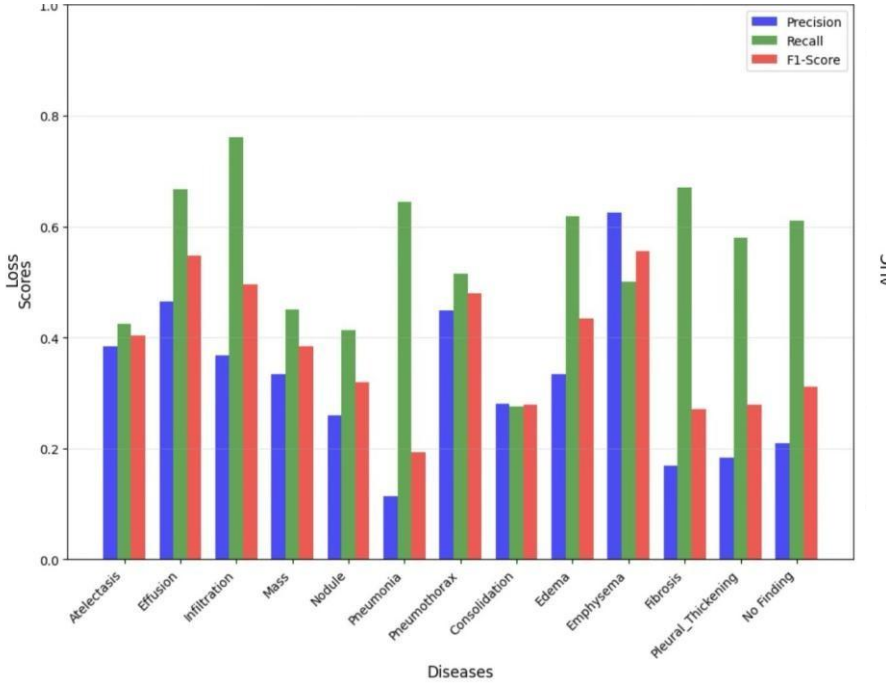


Figure 2: Evaluation metrices

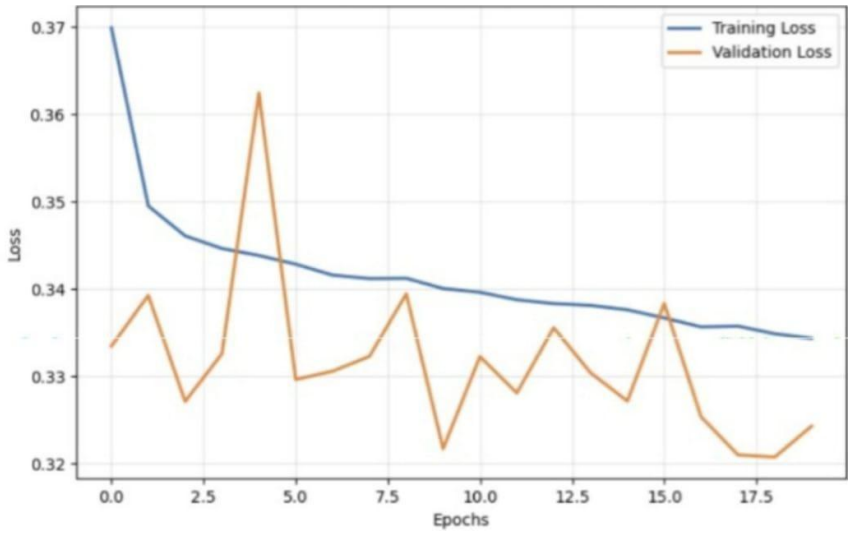


Figure 3: Training vs Validation Loss

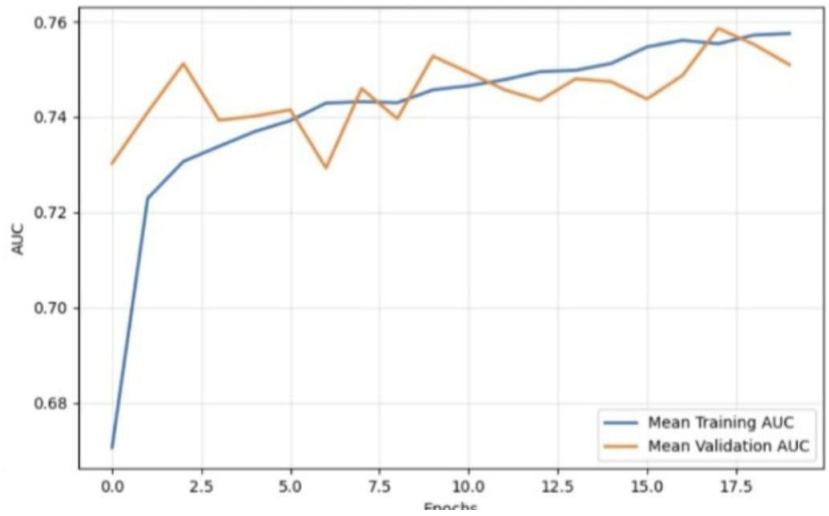


Figure 4: Training vs Validation AUC

4.2. Interpretation

The Grad-CAM technique was used to interpret the behavior of our trained model by generating activation maps that highlight the regions most influential for each disease prediction. These maps were compared with the ground-truth bounding boxes provided in the NIH dataset to assess the model's localization ability. Across multiple test cases—including infiltration, emphysema, and pneumothorax—the model's attention consistently overlapped with the annotated pathological regions, indicating that it successfully learned to focus on clinically relevant structures. The visualizations also showed that the network was able to identify both small, localized abnormalities as well as more diffuse patterns, demonstrating effective and meaningful localization performance for different types of thoracic diseases.

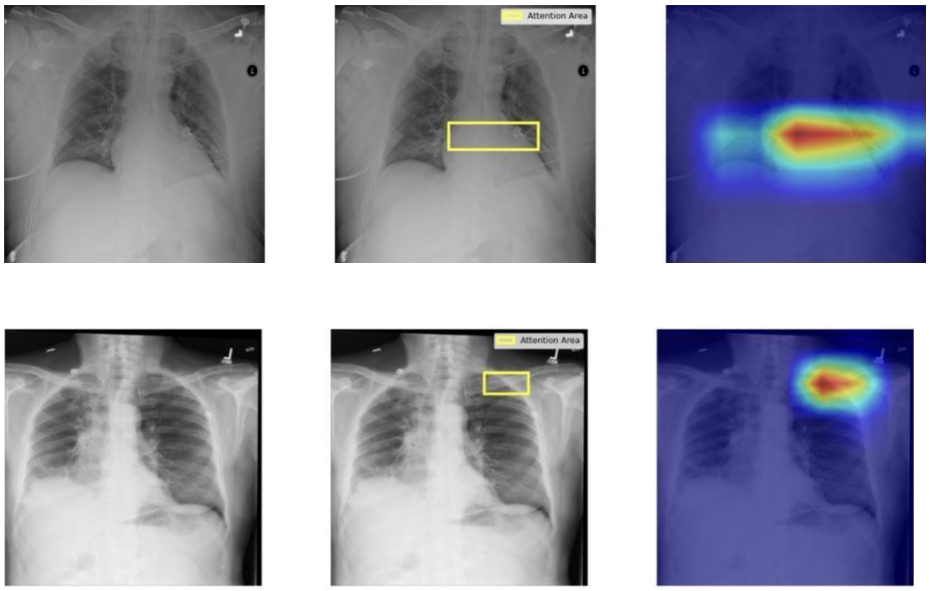


Figure 5: Grad-CAM showing localized pathological areas.

5. Discussion

The experimental results present a nuanced performance profile of our proposed framework in comparison to established methods. While our approach achieved a competitive average AUC of 0.75, it demonstrated variable performance across different pathology classes. Notably, our method excelled in detecting Edema (AUC: 0.855), Emphysema (AUC: 0.872), and Pneumothorax (AUC: 0.819), outperforming

or matching several comparative approaches in these categories. This suggests that our balanced training strategy and EfficientNet-B1 feature extraction effectively captured distinctive patterns for these conditions. However, the model faced challenges with Infiltration (AUC: 0.649), Pneumonia (AUC: 0.662), and Atelectasis (AUC: 0.714), where performance lagged behind other methods. This performance variation highlights the inherent difficulty in detecting certain thoracic pathologies that present with subtle or overlapping radiographic features. The overall results indicate that while our comprehensive approach incorporating data balancing, augmentation, and regularization provides a solid foundation for multi-label classification, there remains significant opportunity for improvement in detecting more challenging conditions. The comparative analysis underscores the need for continued refinement of deep learning approaches for chest X-ray interpretation, particularly for pathologies with complex presentation patterns.

5.1. Comparison

Pathology Label	Yao et al. [12]	Wang et al. [1]	Shen et al. [14]	Guendel et al. [19]	Yan et al. [24]	Baltruschat et al. [25]	Kufel et al. [26]	Ours
Atelectasis	0.733	0.7	0.766	0.767	0.792	0.763	0.761	**0.714**
Effusion	0.806	0.759	0.797	0.828	0.842	0.822	0.775	**0.78**
Infiltration	0.673	0.661	0.751	0.709	0.71	0.694	0.83	**0.649**
Mass	0.777	0.693	0.76	0.821	0.847	0.82	0.843	**0.755**
Nodule	0.724	0.669	0.741	0.758	0.811	0.747	0.727	**0.701**
Pneumonia	0.684	0.658	0.778	0.731	0.74	0.714	0.761	**0.662**
Pneumothorax	0.805	0.799	0.8	0.846	0.876	0.84	0.745	**0.819**
Consolidation	0.711	0.703	0.787	0.745	0.76	0.749	0.775	**0.727**
Edema	0.806	0.805	0.82	0.835	0.848	0.846	0.807	**0.855**
Emphysema	0.842	0.833	0.773	0.895	0.942	0.895	0.83	**0.872**
Fibrosis	0.743	0.786	0.786	0.818	0.833	0.816	0.843	**0.761**
Pleural Thickening	0.724	0.684	0.759	0.761	0.808	0.763	0.727	**0.717**
Average	0.761	0.745	0.775	0.807	0.83	0.806	0.789	**0.75**

Table 3: Comparison with other research papers

This presents a comprehensive comparison of AUC scores across multiple thoracic pathology classification methods, highlighting the performance variations between established approaches and our proposed framework. Our method demonstrates competitive results with an average AUC score of 0.75, positioning it within the spectrum of current state-of-the-art techniques. The model shows particular strength in detecting Edema (0.855), Emphysema (0.872), and Pneumothorax (0.819), achieving performance comparable to or exceeding several reference methods in these categories. However, the results also reveal challenges in classifying Infiltration (0.649), Pneumonia (0.662), and Atelectasis (0.714), where our approach trails behind other meth-

ods. This performance pattern suggests that while our balanced training strategy and EfficientNet-B1 architecture provide robust feature extraction for certain conditions, there remains room for improvement in detecting more subtle pathologies. The comparative analysis underscores the complex nature of multi-label chest X-ray classification and highlights the need for targeted approaches to address specific diagnostic challenges across different thoracic diseases.

6. Conclusion

In conclusion, this study successfully developed and validated a robust deep learning framework for multi-label thoracic disease classification from chest X-rays, demonstrating that the strategic integration of EfficientNet-B1 architecture with comprehensive data balancing, augmentation, and regularization techniques can achieve competitive diagnostic performance. While our approach excelled in detecting several conditions including Edema, Emphysema, and Pneumothorax, it also revealed the persistent challenges in classifying subtle pathologies like Infiltration and Pneumonia, highlighting areas for future improvement. The findings underscore the practical viability of automated computer-aided diagnosis systems while emphasizing the need for continued refinement to handle the complexities of medical image interpretation, ultimately contributing to enhanced clinical decision support and improved patient care outcomes in radiology practice.

7. References

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